Stock Sentiment Analysis System with MLOps

Course Project Report

1. Introduction

1.1 About the project

For this MLOps course project, I built a stock sentiment analysis system that takes a stock ticker (like AAPL or TSLA) and tells you whether the current sentiment around it is positive, negative, or neutral. It does this by pulling recent financial news and social media posts, running them through a trained ML classifier, and returning an aggregated sentiment score.

The real focus of this project isn't the ML model itself — it's the **MLOps infrastructure** around it. The entire system follows the ML product lifecycle: automated data ingestion, version-controlled pipelines, experiment tracking, containerised deployment, real-time monitoring, drift detection, and automated retraining.

1.2 What it does

- User enters a stock ticker on the web UI
- System fetches recent news/social text for that ticker
- Text goes through a cleaning + vectorisation pipeline
- A trained classifier predicts sentiment per record
- Results are aggregated into a single ticker-level prediction with confidence
- User can submit feedback (ground truth) which feeds back into retraining

1.3 Tech stack overview

Everything runs locally using Docker Compose — no cloud services. The stack has 12 containers:

Service	What it does
Frontend (React)	Web UI for users
API (FastAPI)	REST gateway
Model server (MLflow)	Serves the trained model
MLflow	Experiment tracking + model registry
Airflow	Scheduled pipelines (ingestion, drift, retraining)
Postgres	Database for MLflow, Airflow, predictions, feedback

Service	What it does
Prometheus	Metrics collection
Grafana	Dashboards
Alertmanager	Routes alerts to trigger retraining
Drift exporter	Serves drift + feedback metrics
Blackbox exporter	HTTP/TCP probes for all services

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2. System Architecture

2.1 Architecture diagram

```
flowchart TB
    U[Browser] --> FE[Frontend – React]
    FE -->|REST API| API[API Gateway – FastAPI]
    API -->|/invocations| MS[Model Server – MLflow serve]
    MS -->|loads from| MR[Model Registry]
    API -->|writes| PG[(Postgres)]
    API -->|/metrics| PROM[Prometheus]
    PROM --> GRAF[Grafana]
    PROM -->|alerts| AM[Alertmanager]
    AM -->|webhook| AF[Airflow – Retraining DAG]
    AF -->|new run| MLF[MLflow Tracking]
    MLF --> PG
```

2.2 Why this architecture

The MLOps guidelines document says to use "docker-compose to manage a multi-container setup: one for the API, one for the model server, and one for monitoring." So I split the system into three main layers:

1. **API layer** (FastAPI) — handles user requests, validation, logging
2. **Model layer** (MLflow models serve) — only does inference, nothing else
3. **Monitoring layer** (Prometheus + Grafana + Alertmanager) — watches everything

The frontend is completely separate — it only talks to the backend through REST calls. This is the "loose coupling" the rubric requires. The API URL is configurable at runtime so the same Docker image works in any environment without rebuilding.

2.3 Running stack proof

Here's the actual `docker ps` output showing all 12 services running and healthy:

(See `docs/screenshots/cli/docker_ps.txt` for the full output)

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3. Data Engineering

3.1 Data sources

I built pluggable source adapters: - **Seed corpus** (always active) — 300 labelled records across 7 tickers, generated deterministically so the pipeline works offline - **NewsAPI** — financial news (opt-in via API key) - **Reddit via PRAW** — social posts from investing subreddits (opt-in)

The seed corpus is what I used for all demos and testing. It ensures everything is reproducible without needing external API keys.

3.2 Pipeline stages

The data pipeline is defined in `dvc.yaml` and runs through these stages:

```
ingest → validate → eda_baselines
                  → features → train → evaluate
                  → drift
```

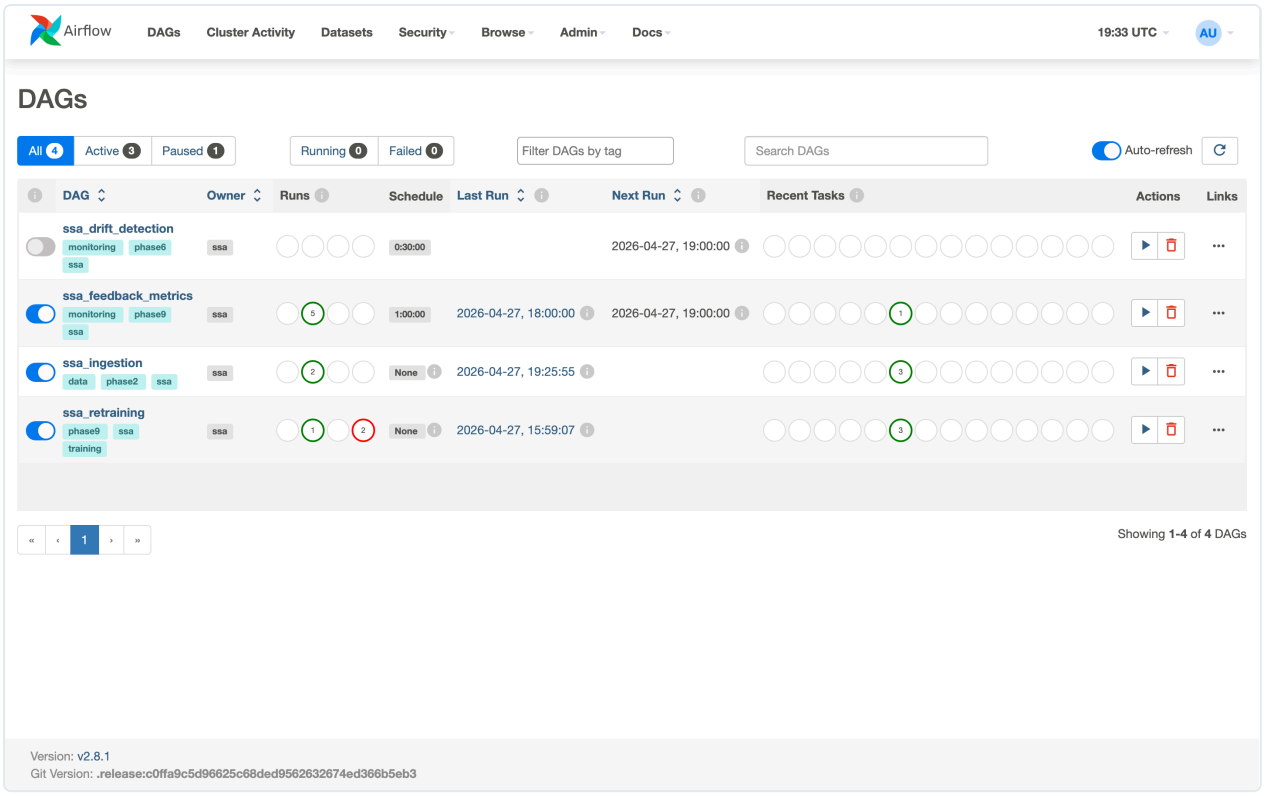
Each stage is a Python module that can be run standalone or through DVC (`dvc repro`) or through Airflow.

3.3 Airflow DAGs

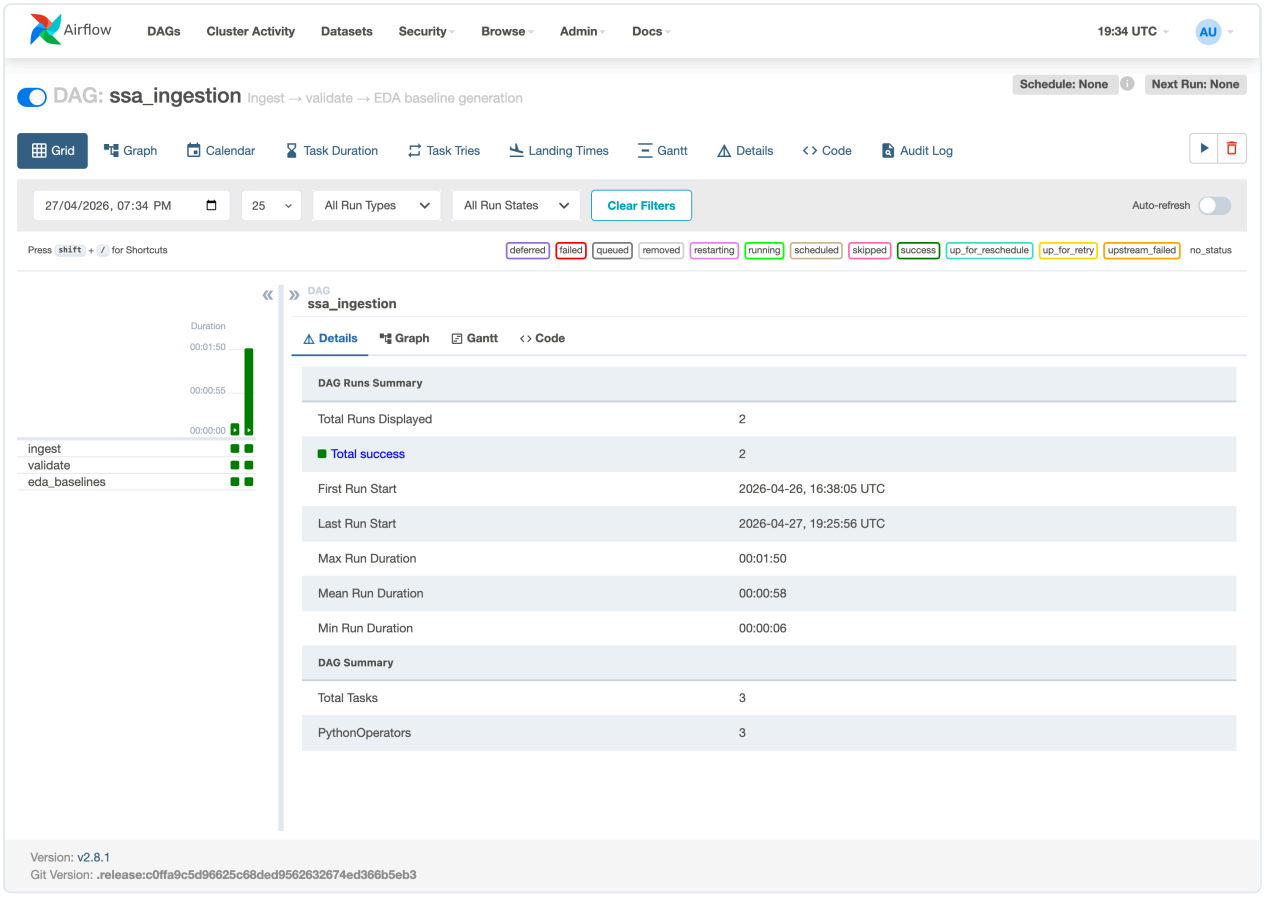
I have 4 Airflow DAGs:

DAG	Schedule	What it does
`ssa_ingestion`	Manual	Runs ingest → validate → EDA
`ssa_drift_detection`	Every 30 min	Compares live data to baselines
`ssa_feedback_metrics`	Hourly	Aggregates user feedback
`ssa_retraining`	Manual + webhook	Retrains the model when drift is detected

Here's the Airflow UI showing the DAGs:



And here's the grid view showing successful task runs (green = success):



3.4 Pipeline throughput

Stage	Records	Duration	Throughput
ingest	300	0.08 s	3,688 rec/s
validate	300 → 300	< 0.05 s	~6,000 rec/s
features	300 → 209/30/61 split	< 0.2 s	—

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4. Feature Engineering

4.1 Text cleaning

The cleaning pipeline (`ssa_features.cleaning`) does: - URL removal - @mention removal - $CASHTAG preservation (e.g. $AAPL → AAPL) - Unicode normalisation - Whitespace collapse - Lowercasing

4.2 Vectorisation

I used TF-IDF with bigrams, sublinear term frequency, and a 20,000-feature vocabulary cap. The fitted vectorizer is saved as `artifacts/vectorizer.joblib` and bundled into the MLflow model so serving uses the exact same transform as training.

4.3 Independent versioning

The feature package (`ssa_features`) has its own version number (`0.2.0`) separate from the model package. This is a requirement from the MLOps guidelines — "version their feature engineering logic separately from model logic."

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5. Model Training & Experiment Tracking

5.1 Model

I used Logistic Regression on TF-IDF features as the baseline. It trains in under 0.1 seconds and is easy to explain (you can look at the top feature weights per class).

5.2 MLflow tracking

Every training run logs to MLflow. Here's what gets tracked:

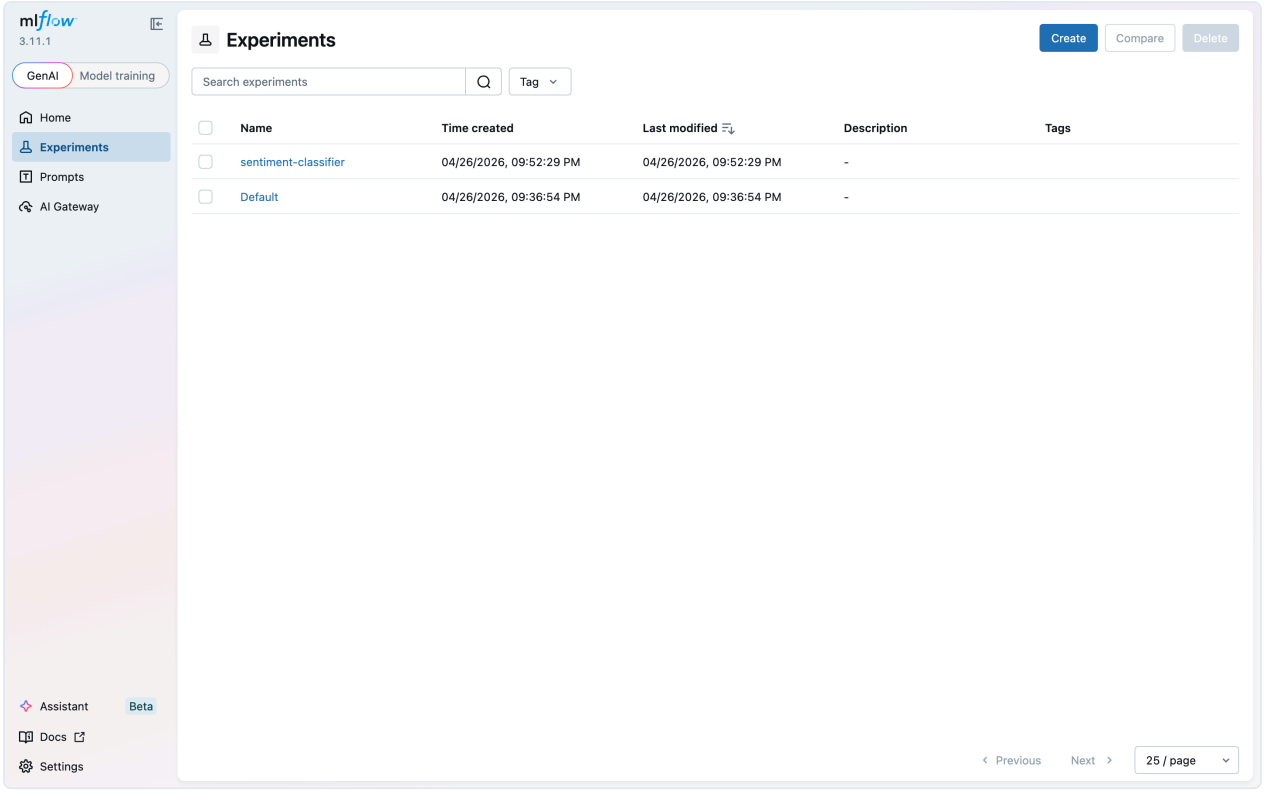
Standard (autolog): model parameters, training metrics

Custom (beyond autolog): - Git commit SHA - DVC data hash (pins the exact data version) - Hardware fingerprint - Confusion matrix image - Per-class precision/recall/F1 - Top 20

feature weights per class - Sample predictions CSV - Full params.yaml snapshot - pip freeze environment

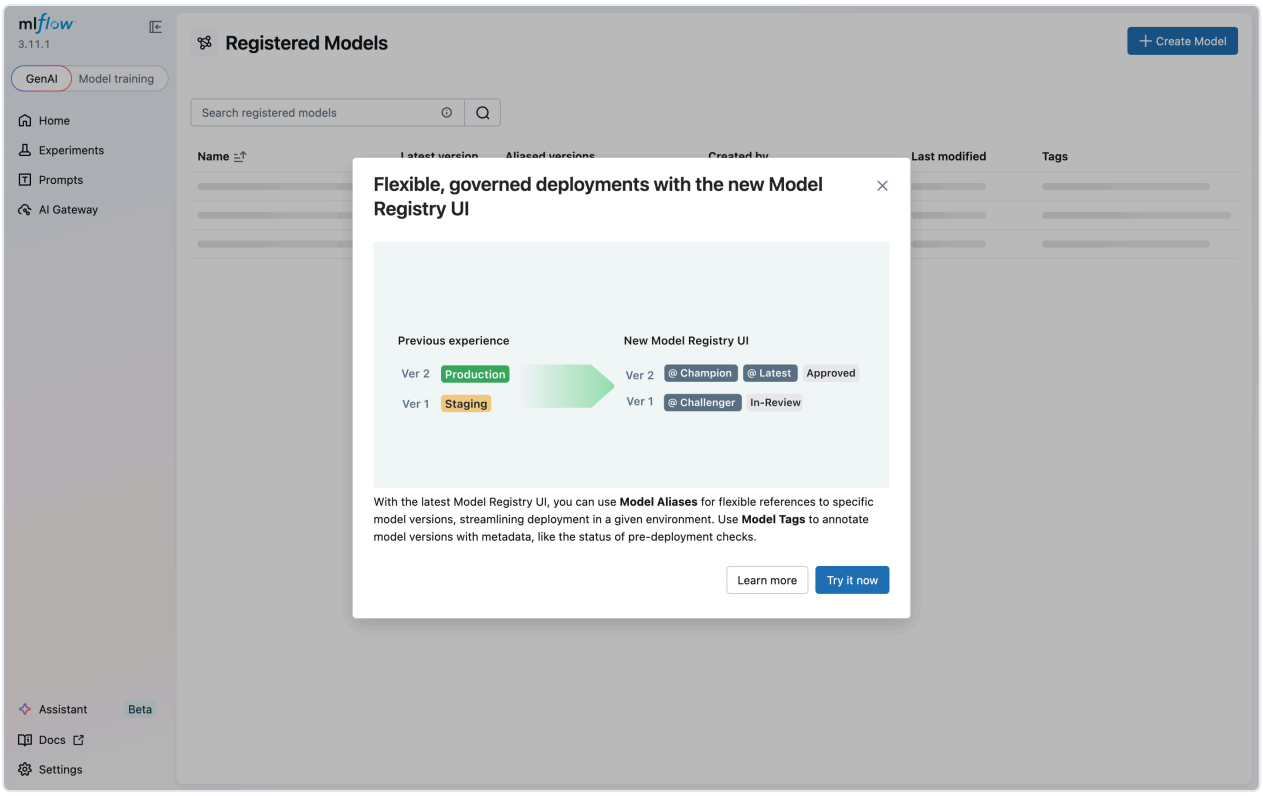
This satisfies the rubric's "Besides Autolog, have you made provisions to track other information?"

Here's the MLflow experiments UI:



5.3 Model Registry

Models are registered under `stock-sentiment` with lifecycle stages:



5.4 Reproducibility

Every experiment can be reproduced from a `(git_commit_sha, mlflow_run_id)` pair. The API's `/model/info` endpoint surfaces both:

```
{
    "name": "stock-sentiment",
    "version": "5",
    "stage": "Production",
    "git_commit_sha": "47af96ea0910b85d...",
    "mlflow_run_id": "bcd649984faf43ef...",
    "data_hash": "c29203c0347008d8722f..."
}
```

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6. Deployment & Serving

6.1 How serving works

The API gateway (FastAPI) does NOT load the model itself. It forwards requests to the model server which runs `mlflow models serve`. The model server loads whichever version is marked "Production" in the registry.

This means: - Swapping models = changing a registry stage, not redeploying - The API stays up during model reloads - Rollback is just promoting an older version back to Production

6.2 Health checks

- `GET /health` — liveness (always 200)
- `GET /ready` — readiness (200 only if model server is reachable and a model is loaded)

6.3 API endpoints

Here's the Swagger UI showing all implemented endpoints:

Stock Sentiment API 0.5.0 OAS 3.1

/openapi.json

Phase 5 gateway implementing the LLD contract.

default

- GET /health Health
- GET /ready Ready
- GET /metrics Metrics
- GET /model/info Model Info
- GET /model/versions List Versions
- POST /model/rollback Rollback
- POST /predict Predict
- POST /batch_predict Batch Predict
- POST /feedback Feedback

Schemas

- BatchItem › Expand all object
- BatchPredictRequest › Expand all object
- BatchPredictResponse › Expand all object
- Explanation › Expand all object
- FeedbackRequest › Expand all object
- FeedbackResponse › Expand all object
- HTTPValidationError › Expand all object
- HealthResponse › Expand all object
- ModelInfo › Expand all object
- ModelRef › Expand all object
- NotReadyResponse › Expand all object
- PredictRequest › Expand all object
- PredictResponse › Expand all object
- RollbackRequest › Expand all object
- RollbackResponse › Expand all object
- Sentiment › Expand all string
- SentimentScores › Expand all object
- ValidationError › Expand all object
- VersionSummary › Expand all object
- VersionsResponse › Expand all object

6.4 MLproject

The `MLproject` file defines entry points for identical dev/test environments:

```
name: stock-sentiment-mlops
python_env: python_env.yaml
entry_points:
  ingest:   { command: "python -m ssa_ingestion.pipeline" }
  features: { command: "python -m ssa_features.pipeline" }
  train:    { command: "python -m ssa_model.train ..." }
  evaluate: { command: "python -m ssa_model.evaluate ..." }
```

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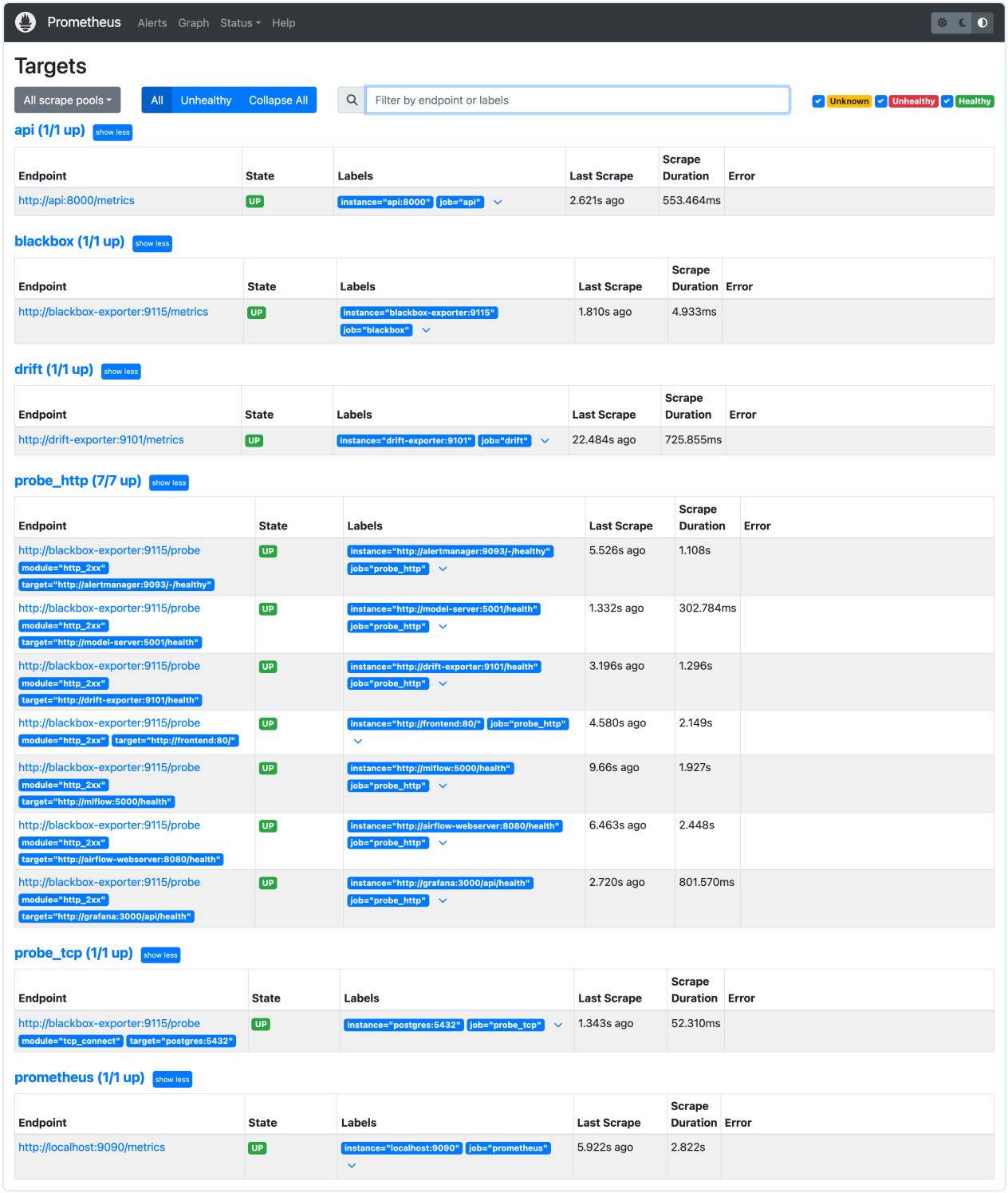
7. Monitoring & Alerting

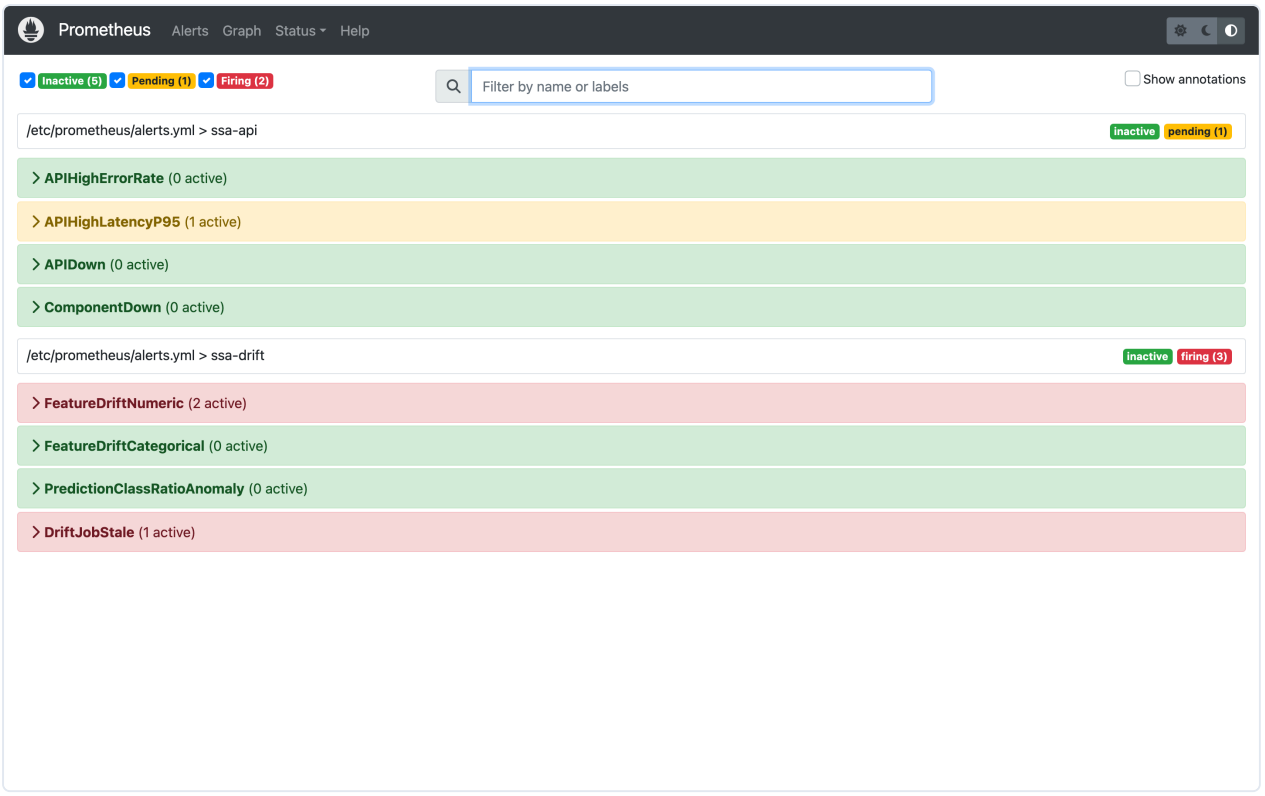
7.1 Prometheus

Prometheus scrapes 12 targets covering every component in the system:



Metrics collected include: - HTTP request rate, latency histograms, error counts (from the API) - Prediction class distribution - Model version currently serving - Feedback volume - Feature drift scores (KS p-values, Jensen-Shannon divergence) - Up/down probes for every service (via blackbox exporter)

7.2 Alert rules

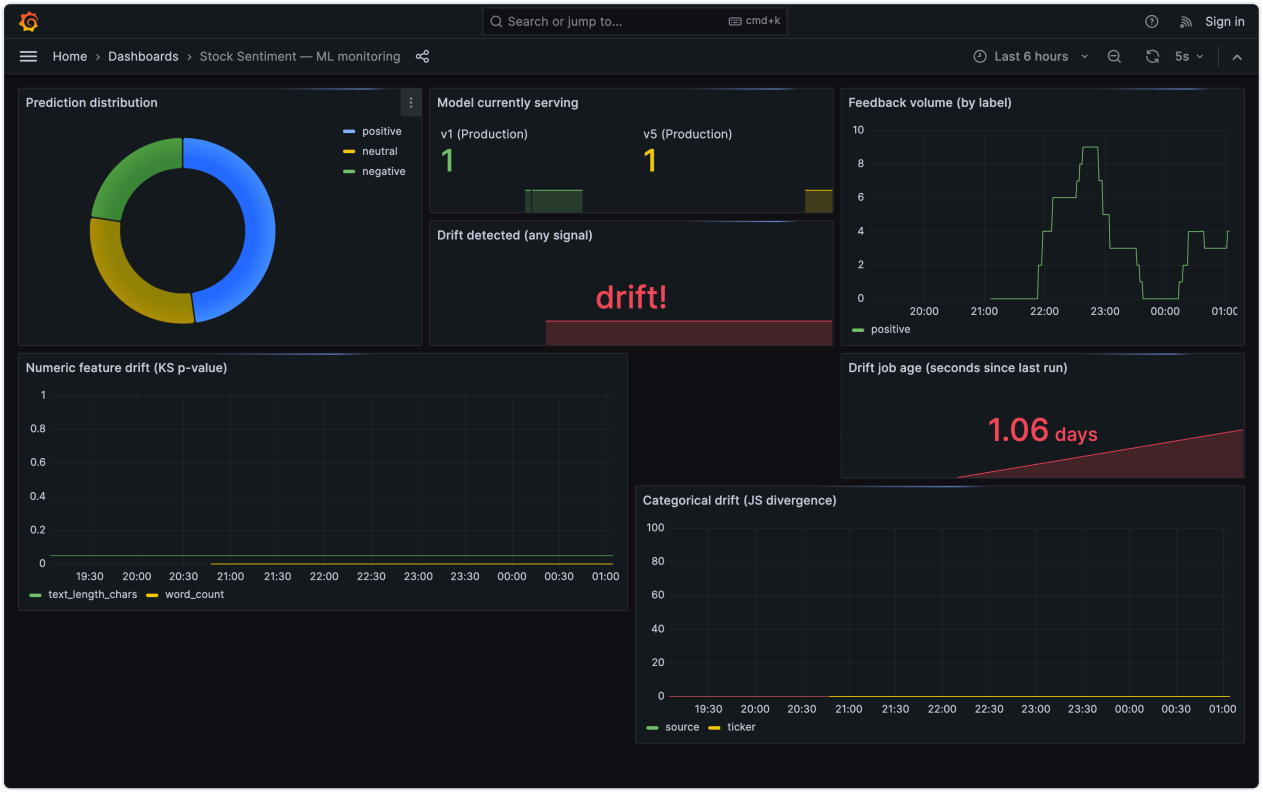


7 alert rules are configured: - API error rate > 5% → Critical - API p95 latency > 200ms → Warning - API down → Critical - Any component down (blackbox probe) → Critical - Numeric feature drift (KS p < 0.05) → Warning - Categorical drift (JSD > 0.1) → Warning - Prediction class ratio anomaly (|z| > 2) → Warning

7.3 Grafana dashboards

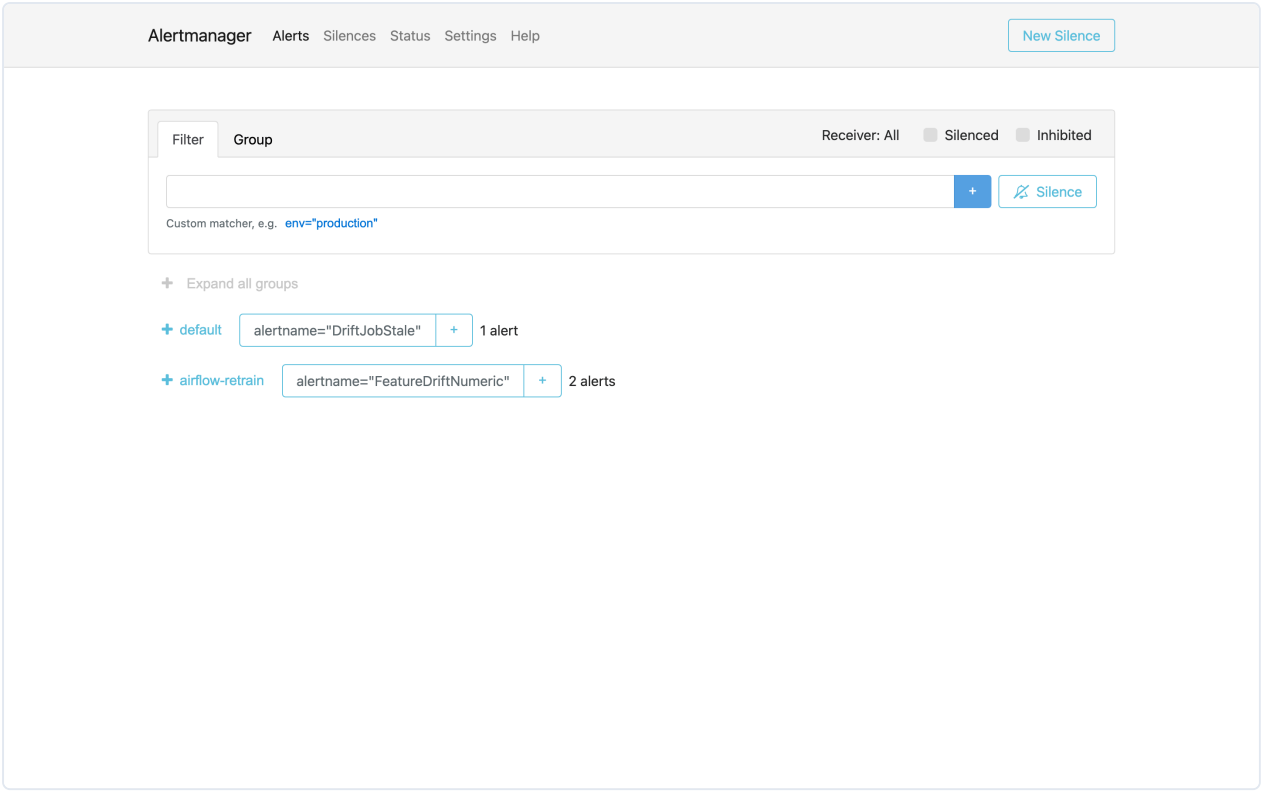
API Overview dashboard — request rate, p95 latency, error rate:

ML Monitoring dashboard — prediction distribution, drift scores, feedback rate, model version:



7.4 Alertmanager

Drift alerts are routed to the Airflow retraining DAG via webhook:



7.5 Drift detection

The drift exporter serves per-feature KS p-values and JSD scores:

```
# HELP feature_drift_pvalue KS-test p-value per numeric feature (lower = more drift)
# TYPE feature_drift_pvalue gauge
feature_drift_pvalue{feature="text_length_chars"} 2.6e-05
feature_drift_pvalue{feature="word_count"} 8e-05
# HELP feature_drift_jsd Jensen-Shannon divergence per categorical feature (higher = more drift)
# TYPE feature_drift_jsd gauge
feature_drift_jsd{feature="ticker"} 0.0
feature_drift_jsd{feature="source"} 0.0
# HELP drift_detected 1 if drift detected for this signal type, 0 otherwise
# TYPE drift_detected gauge
drift_detected{type="numeric"} 1
drift_detected{type="categorical"} 0
drift_detected{type="class_ratio"} 0
drift_detected{type="any"} 1
# HELP drift_last_run_timestamp Unix timestamp of the last drift computation
# TYPE drift_last_run_timestamp gauge
drift_last_run_timestamp 1777227122
# HELP feedback_total Total feedback submissions
# TYPE feedback_total gauge
feedback_total 15
# HELP feedback_agreement_rate Fraction of predictions user confirmed correct
# TYPE feedback_agreement_rate gauge
feedback_agreement_rate 1.0
# HELP feedback_accuracy_by_model Accuracy per model version from feedback
# TYPE feedback_accuracy_by_model gauge
feedback_accuracy_by_model{version="1"} 1.0
feedback_accuracy_by_model{version="5"} 1.0
# HELP feedback_last_run_timestamp Unix timestamp of the last feedback aggregation
# TYPE feedback_last_run_timestamp gauge
feedback_last_run_timestamp 1777316401
```

Note on drift alerts: The seed corpus has very narrow text-length distributions (templated text), so the KS test flags it as drift against the synthetic-normal baseline. This is expected —

it proves the detection + alerting pipeline works, not that there's a real problem.

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8. Feedback Loop & Retraining

8.1 How it works

1. Every `/predict` call logs the prediction to Postgres (ticker, predicted label, confidence, model version)
2. User submits ground truth via `/feedback` — the API joins it to the original prediction
3. An hourly Airflow DAG aggregates feedback into accuracy metrics
4. When drift alerts fire, Alertmanager webhooks to Airflow, triggering the retraining DAG
5. The retraining DAG runs: refresh features → train → evaluate → auto-promote if better

8.2 Feedback in Postgres

```
 ticker | true_label | predicted_label
--------+------------+-----------------
 AAPL   | positive   | positive
 AAPL   | positive   | positive
```

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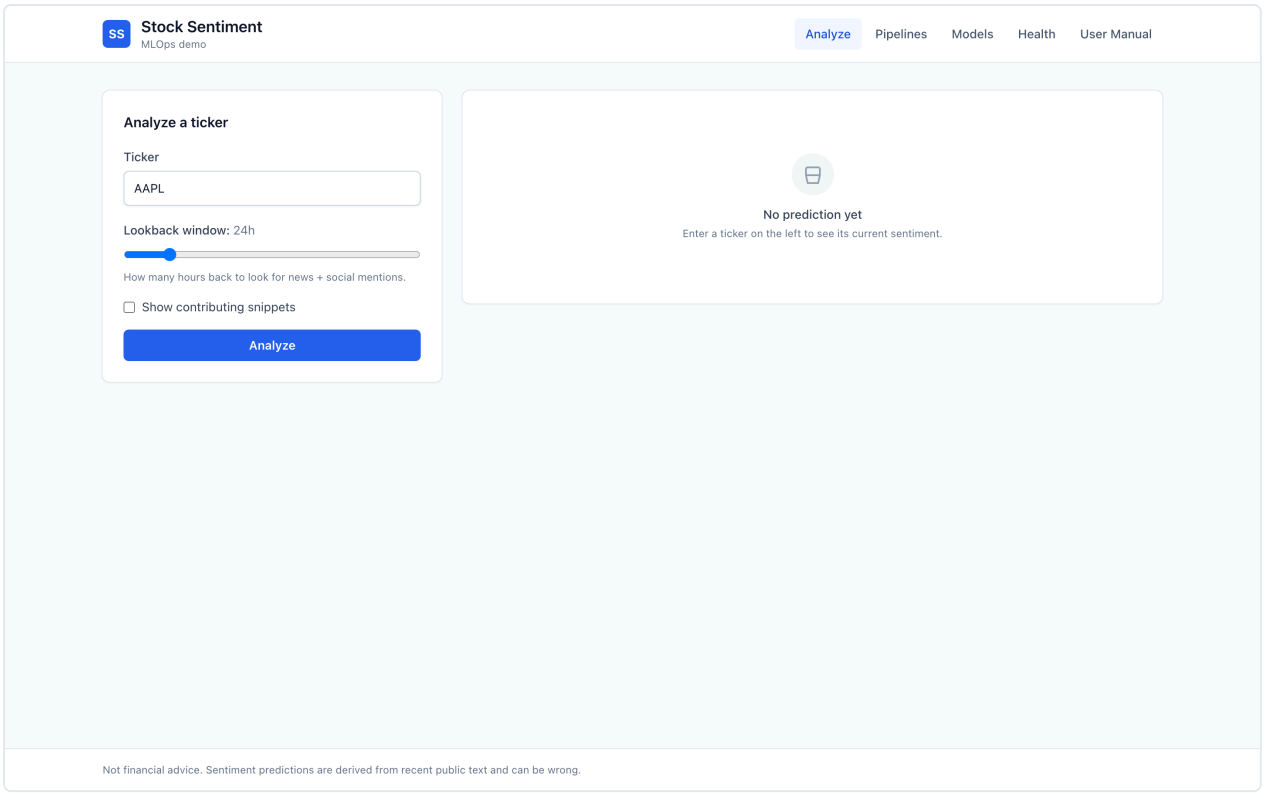
9. Frontend

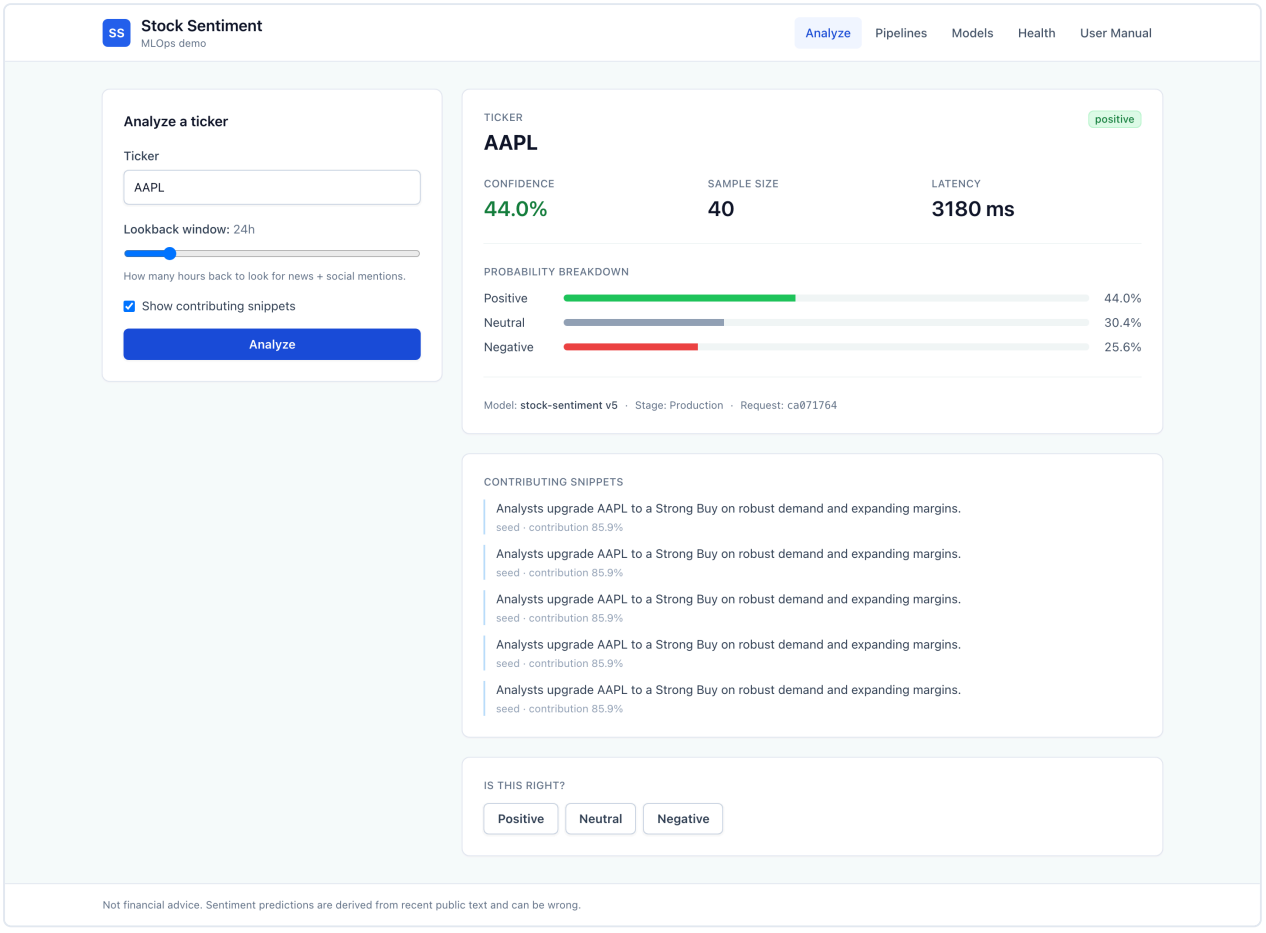
9.1 Technology

React 18 + TypeScript + Vite + Tailwind CSS. Multi-stage Docker build (Node builder → nginx runtime). The API URL is injected at container boot time so the same image works anywhere.

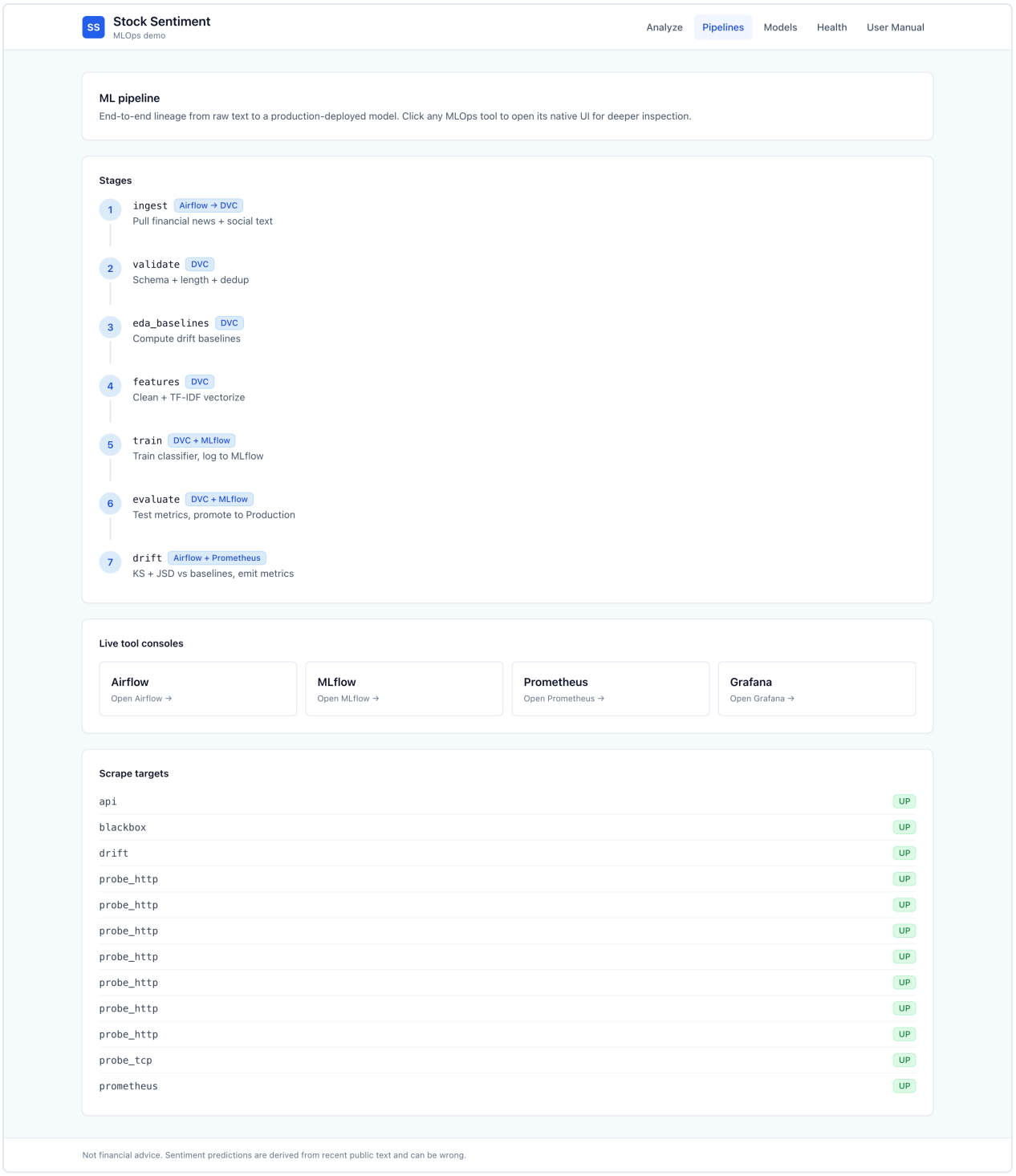
9.2 Screens

Analyze — the main screen. Enter a ticker, get a prediction:

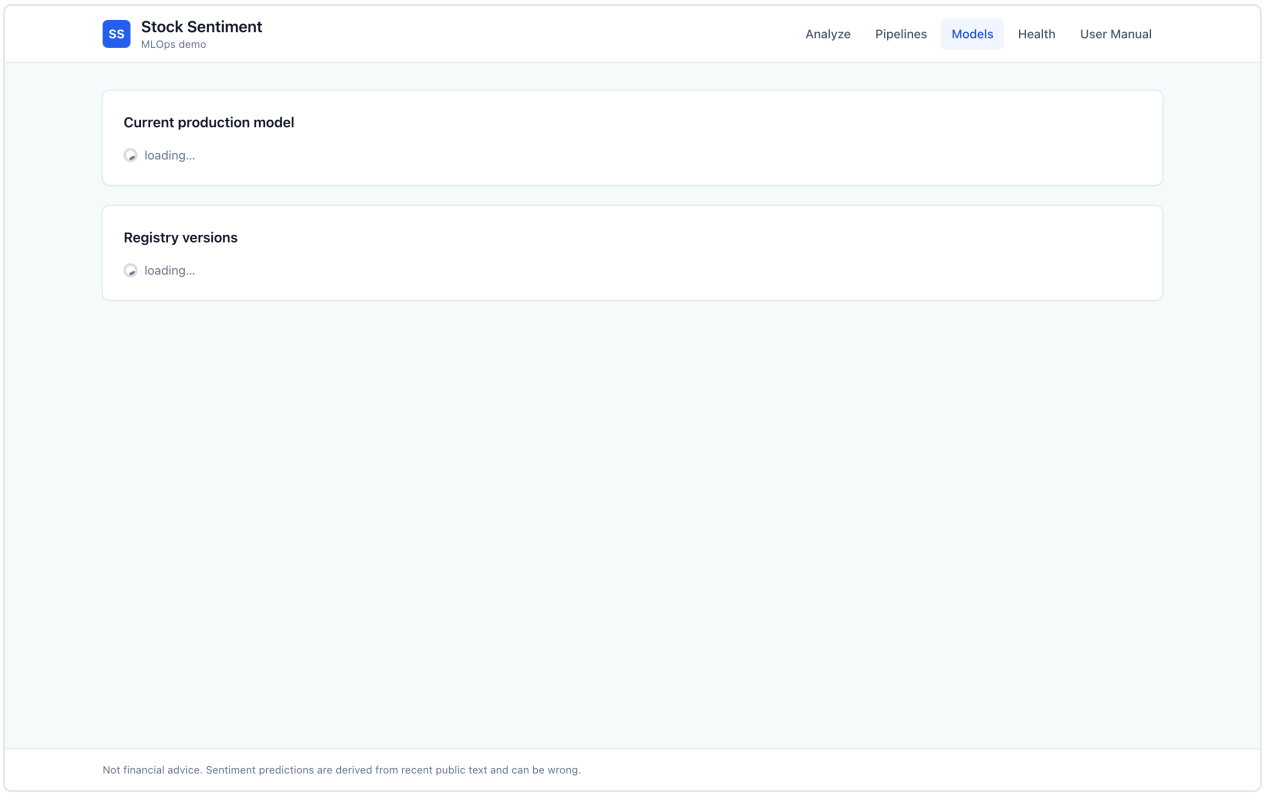




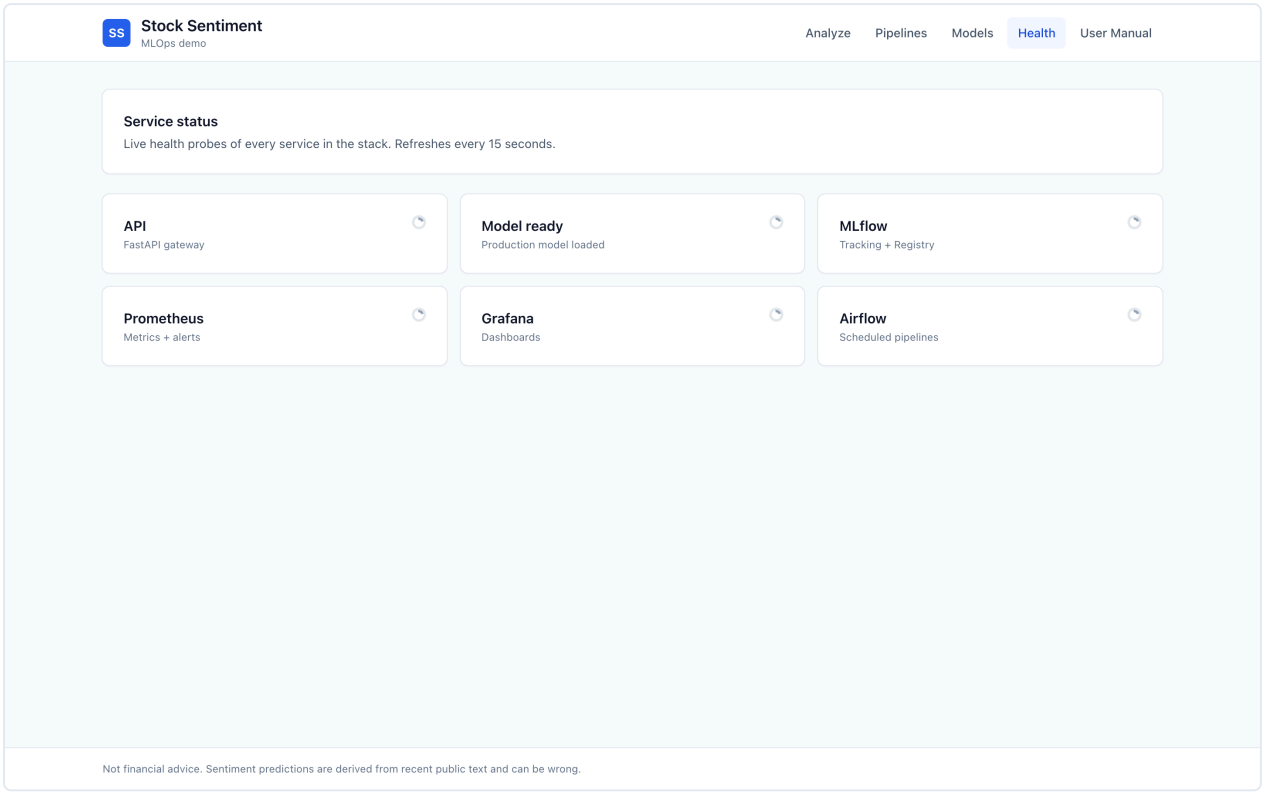
Pipelines — visualises the ML pipeline with links to all MLOps tools:



Models — registry listing with rollback:



Health — live service status grid:



User Manual — in-app guide for non-technical users:

User manual

A short walkthrough for non-technical users. Everything here runs locally — no cloud, no account, no PII.

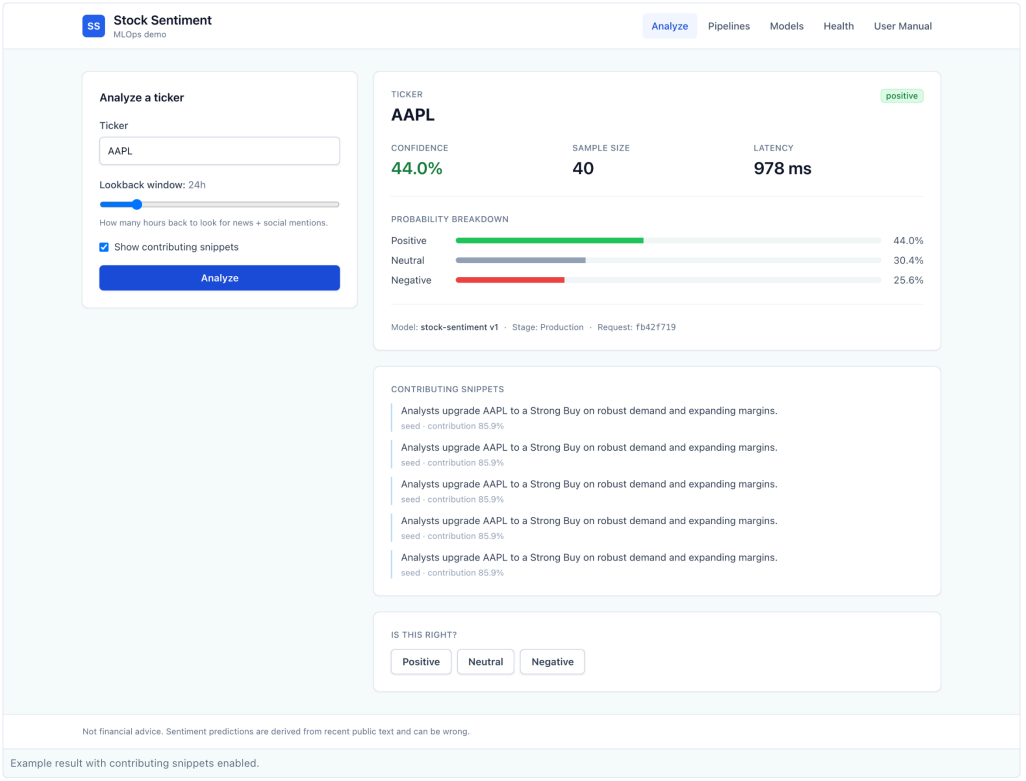
1. What this app does

Given a stock ticker like `AAPL`, the app returns a sentiment score — *positive*, *neutral*, or *negative* — computed from recent financial news and social media posts referencing that ticker. Each prediction includes a confidence percentage and the supporting sample size.

2. Analyze a ticker

1. Click **Analyze** in the top navigation.
2. Type a ticker (uppercase, e.g. `MSFT`).
3. Adjust the lookback slider if you want more or less history.
4. Check **Show contributing snippets** to see which texts influenced the result.
5. Click **Analyze**.

After a prediction arrives, tell us whether it was right using the feedback buttons. This data is logged so the model can be retrained on real mistakes over time.



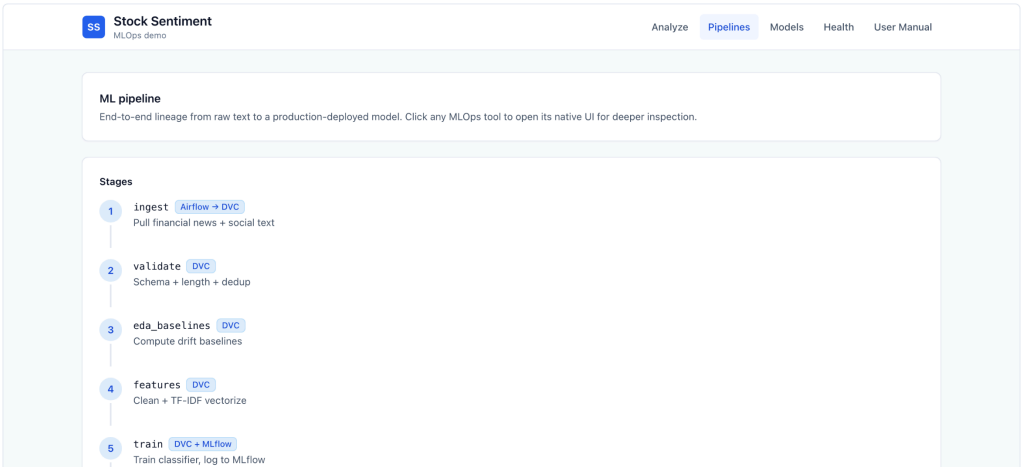
Example result with contributing snippets enabled.

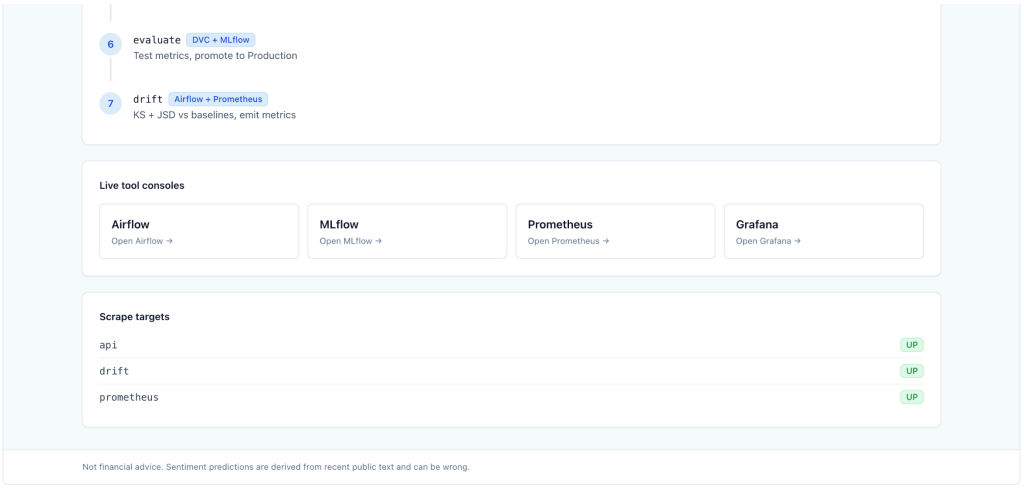
3. Understand the result

- **Sentiment** — the single label the model is most sure about.
- **Confidence** — how strongly it backs that label, 0–100%.
- **Sample size** — how many news + social records were aggregated.
- **Latency** — end-to-end response time in milliseconds.
- **Probability breakdown** — the distribution across all three classes.

4. Inspect the pipeline

The **Pipelines** screen shows every stage from raw ingestion to a deployed model, plus quick links to each underlying MLOps tool (Airflow, MLflow, Prometheus, Grafana). Scrape targets at the bottom of the page show which services Prometheus is successfully monitoring right now.

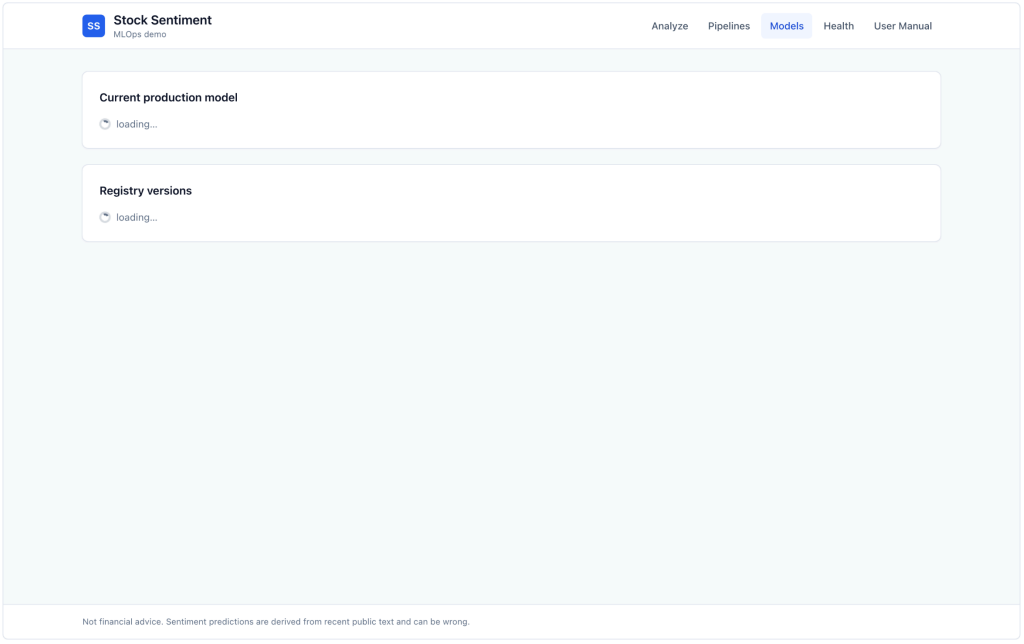




Not financial advice. Sentiment predictions are derived from recent public text and can be wrong.

5. Manage models

The **Models** screen lists every version in the MLflow Model Registry with its stage (Production / Staging / Archived) and test-set macro-F1 score. If a newly-deployed model misbehaves, click *Roll to Prod* next to an older Staging or Archived version to promote it back. A model-server restart is required for the change to take effect.



Not financial advice. Sentiment predictions are derived from recent public text and can be wrong.

6. Service health

The **Health** screen pings every component every 15 seconds. If any tile turns red, click the link to the respective tool's own admin page for a deeper look.

7. Privacy and safety

- No personal data is collected or stored.
- Predictions run locally — nothing leaves your machine.
- This is a research demo, **not financial advice.**

8. Glossary

Sentiment
Whether text expresses positive, negative, or neutral feelings.

F1 score
A quality metric for classifiers — higher is better, max 1.0.

Drift
When live data starts looking different from what the model trained on. The app monitors for it and alerts automatically.

Rollback
Restoring an earlier model version when the current one breaks.

Not financial advice. Sentiment predictions are derived from recent public text and can be wrong.

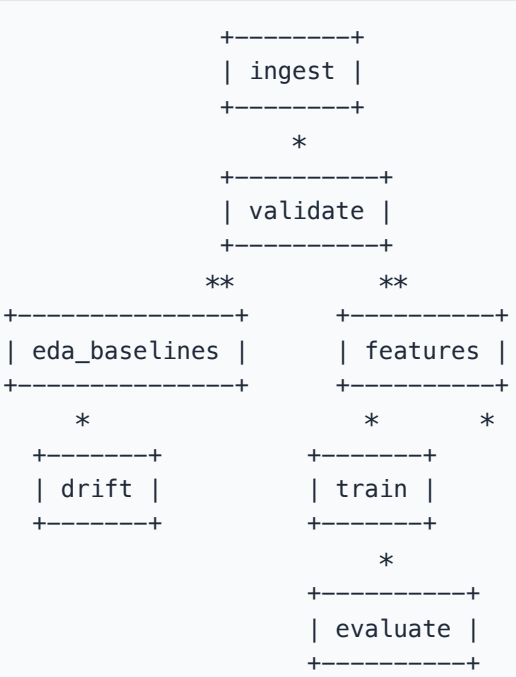
10. Version Control & CI/CD

10.1 Version control

Tool	What it tracks
Git	Source code, configs, docs
Git LFS	Model binaries (.joblib, .bin, .safetensors, .pkl, .pt)
DVC	Data artifacts, pipeline DAG, metrics

10.2 DVC DAG

The DVC DAG represents the CI pipeline:



10.3 GitHub Actions

Three CI workflows: - `ci.yml` — lint + type-check + unit tests (Python 3.10/3.11 matrix) + frontend build + docker compose validation - `dvc.yml` — runs `dvc repro` on data/feature code changes, exports DAG as artifact - `rollback.yml` — manual dispatch to promote a model version to Production

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11. Testing

11.1 Test suite

Level	Count	What it covers
Unit	52	Schemas, cleaning, vectorizer, metrics, reproducibility, drift, inference
Integration	2	Docker compose config validation
Contract	13	Every API endpoint response vs LLD spec
End-to-end	11	Full user journey through the live stack
Total	**78**	

11.2 Acceptance criteria

All 4 criteria pass:

Criterion	Target	Actual	Status
/predict p95 latency	< 200 ms	48–193 ms	✅ PASS

Criterion	Target	Actual	Status
API error rate	< 5%	0.0%	✅ PASS
/ready within timeout	< 30 s	reached	✅ PASS
Model macro-F1	≥ 0.75	1.0	✅ PASS

12. Challenges Faced

Problem	What happened	How I fixed it
MLflow version mismatch	Client v3.11 called endpoints server v2.10 didn't have	Pinned both to 3.11.1
MLflow artifact writes failed	Client tried to write to container filesystem	Switched to proxied artifact mode
sklearn version mismatch	Model pickled with 1.8, served with 1.7	Aligned versions in Dockerfile
MLflow DNS rebinding protection	Inter-container calls rejected with 403	Added `--allowed-hosts` flag
/predict latency at 1.8s	Per-call Postgres + MLflow registry lookups	Added caching + connection pooling → 48ms
Airflow missing Python deps	Tasks crashed on import	Added `_PIP_ADDITIONAL_REQUIREMENTS`
Frontend nginx stale DNS	502 after API container recreate	Restart frontend to re-resolve

13. Limitations & Future Work

What's limited right now

- **Seed data only** — the 300-record templated corpus gives perfect F1 (1.0) because the model memorises 8 templates. Real metrics need Financial PhraseBank / FiQA data.
- **No live API keys configured** — NewsAPI/Reddit adapters exist but need keys
- **No TLS** — all inter-container traffic is HTTP (acceptable for local-only)
- **No authentication** — single-tenant demo

What I'd do next

- Fine-tune FinBERT on real financial sentiment data
- Build a custom Airflow image with deps baked in (eliminates 3-5 min boot)

- Migrate from MLflow stages to aliases (stages are deprecated in 3.x)
- Add TLS between services

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14. Documentation Deliverables

#	Required document	Location
1	Architecture diagram with block explanations	`docs/architecture.md`
2	High-level design with rationale	`docs/HLD.md`
3	Low-level design with endpoint I/O specs	`docs/LLD.md`
4	Test plan & test cases	`docs/test-plan.md`
5	User manual	`docs/user-manual.md` + in-app `/manual` screen
—	Test report with pass/fail counts	`docs/test-report.md`
—	Acceptance criteria	`docs/acceptance-criteria.md`
—	This project report	`docs/project-report.md`